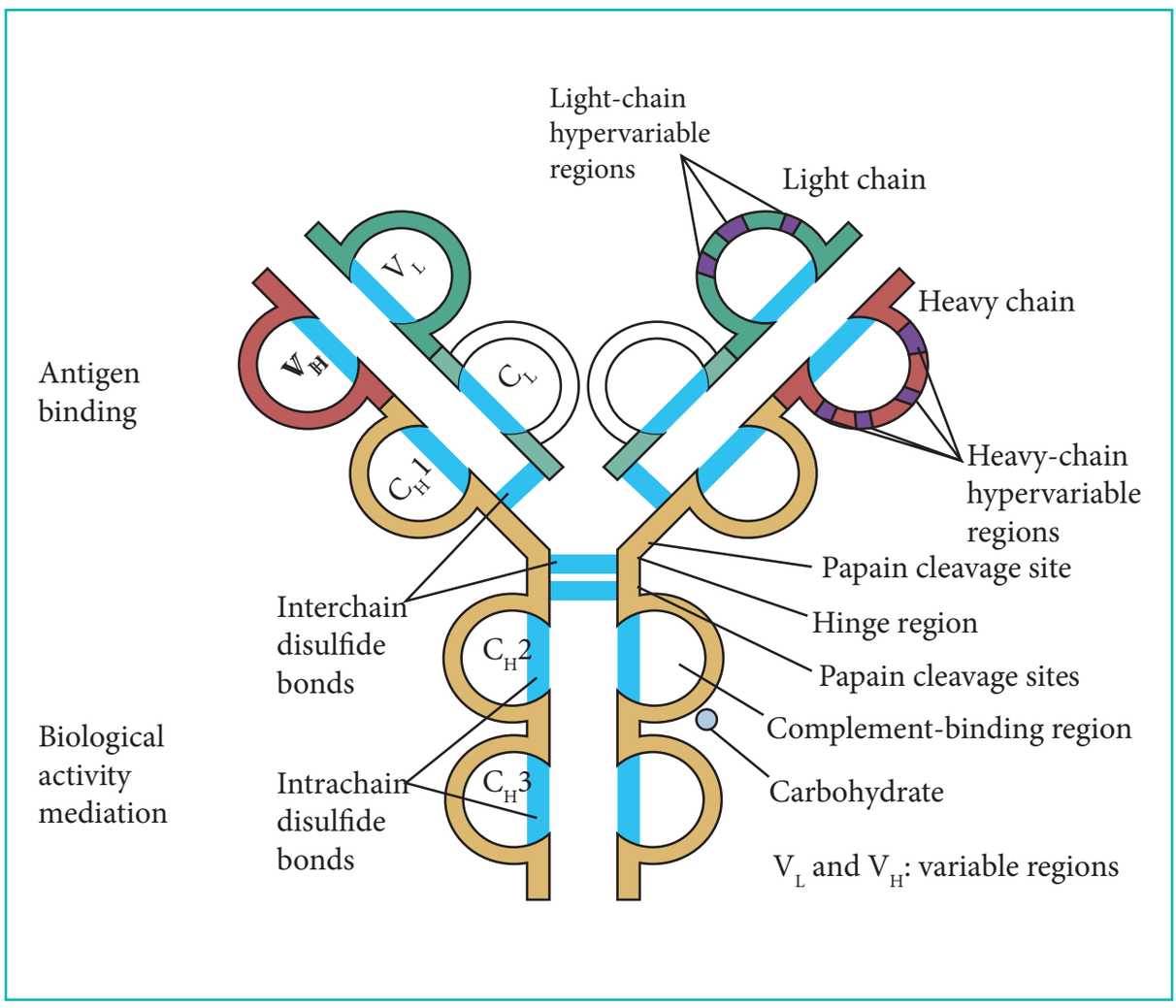
length of the constant region is approximately 330 amino acids for gamma (IgG), delta (IgD) and alpha (IgA) and 440 amino acids for mu (IgM) and epsilon (IgE). In humans there are two subclasses of alpha heavy chains and four subclasses of gamma chains .

Light chain domain consisting of 110 amino acids was found to have varying amino acid sequences in different antibodies for different antigens at amino terminal end. This region was called as the variable (V) region. The carboxy terminal half of the molecule, is called the constant (C) region. There are two types of light chains , kappa and Lambda . In humans 60% of the light chains are kappa and 40% are lambda. A single antibody molecule contains either kappa chains or lambda chains, but never both.

The gamma, delta, and alpha heavy chains contain an extended peptide sequence between the C_{H1} and C_{H2} domains. This region, called the hinge region**,** is rich in proline residues and is flexible.

A paratope (antigen-binding site) is a part of an antibody which recognizes and binds to an antigen. It is a small region of the antibody's variable region, and contains parts of the antibody's V_H domain of heavy and V_L domain of light chains. This region is also called as complementarity determining region (CDR).



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